

HIKET — the storyline, v2

skeleton for discussion: headers, one-line arguments, figures

Version of September 2, 2026. Numbers keyed to RUN_IDs 20260831_1624* (arm B). Deliberately not prose — the point is to argue about the ARC.

How to read this. Every block is a claim we intend to make, its evidence, and nothing else. ▶ **name** marks a figure that exists; ★ **TO BUILD: name** marks one that does not yet. Green boxes are *tests we could run*; amber boxes are *things that could sink the claim*.

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1 The arc in one paragraph

Finnish forest soils are accumulating carbon and the accumulation is beginning to level off. A model started at steady state cannot show that, because a steady state is already the end of it. Started instead from a calibrated past, all six models reach the observed levels and two of them reproduce the rate exactly — but doing so requires *more litter input and faster turnover* than the published parameterisations carry. That displacement is the finding, and it is not free: it should change how the soil responds to a warming climate. **The new work is to test whether it does.**

Part I

Act I — Opening

2 A sink caught saturating

2.0.0.1 The observed dynamic. On the balanced plot set the mean stock runs $64.5 \rightarrow 71.5 \rightarrow 73.6 \text{ tC ha}^{-1}$ across the three campaigns: +7.0 then +2.1. Accumulation, bending over. ► [F3_mean_soc](#)

2.0.0.2 The rate is what an inventory reports. $+0.259 \pm 0.040 \text{ tC ha}^{-1} \text{ yr}^{-1}$ over the full span. ► [F5_stock_change](#)

2.0.0.3 Why it is politically live. The Finnish GHG inventory turns on whether this sink persists or saturates. ► [F10b_forest_history](#)

3 The convention that cannot see it

3.0.0.1 Steady-state initialisation is near-universal and rarely examined. Lehtonen 2016, Palosuo 2008, Peltoniemi 2004 — our own group’s lineage, to be framed as evolving, not attacked.

3.0.0.2 An equilibrium start is already at the end. It cannot produce a rise, so it cannot produce a rise that is levelling off.

Part II

Act II — Challenge

4 To show the saturation, the model must remember it was rising

4.0.0.1 The soil carries a memory the model is told to forget. Slow pools integrate a century of inputs; setting them to equilibrium erases exactly the transient the data show.

4.0.0.2 Managed landscapes are rarely at equilibrium. Finland is the vivid case, not a special one: exploitation, then a century of managed regrowth. ► [F10b_forest_history](#)

Part III

Act III — Action

5 Transient initialisation under a fair frame

5.0.0.1 The unknown start becomes a calibrated parameter. Each draw spins up from 1917; σ_{init} is the ratio of 1917 to 1985 litter input. Uncertainty propagates rather than being patched. ► [F4_initialization](#)

5.0.0.2 Fair means equal *information*, not equal parameter count. Yasso does not fit its decomposition rates, so neither do the simple models: rates externally anchored (ICBM), fractions free. Otherwise structure and calibration asymmetry are confounded.

5.0.0.3 One likelihood, and it does not pretend the plots are independent. Log-normal with a fixed correlated error structure (region, plot, campaign), total $\sigma = 0.800$ partitioned, not added. ★ TO BUILD: [schematic of the variance partition — optional](#)

Part IV

Act IV — Resolution, part 1: it works

6 The models reach the observations

6.0.0.1 Levels. No model sits more than 3.5 tC ha^{-1} from any campaign mean. The old “ 26 tC ha^{-1} too high at 1985” caveat is gone. ► [F3_mean_soc](#)

6.0.0.2 The rate. Against observed $+0.259$: Yasso07 $+0.258$, Yasso15 $+0.250$. TP2/TP3 about a third strong; SP1/Yasso20 about half weak. The ensemble brackets the answer. ► [F5_stock_change](#)

6.0.0.3 The three-way split — new, and better than “all six work”. Yasso07/Yasso15 rise then flatten (*saturation reproduced*); TP2/TP3 rise and keep rising (*no saturation*); SP1/Yasso20 overshoot then decline. Structure is legible in the shape. ★ TO BUILD: [F3b — trajectory shape, three groups, explicitly grouped](#)

The 2006–2024 interval discriminates far less than it looks: observed $+0.117 \pm 0.066$, CI includes zero. Do not lean on “two models get the sign wrong” over eighteen years.

Part V

Act V — Thread A, restated

7 Fit does not discriminate structure; dynamics do

This replaces “does complexity improve the answer?”. The old answer (“no, and the most complex fits worst”) is no longer true — Yasso15 has the best calibration R^2 and the best holdout RMSE. The defensible and more useful claim is the one below.

7.1 In-sample, structure is invisible

7.1.0.1 Skill does not climb the pool ladder. Calibration R^2 0.016–0.021; holdout $R^2 \leq 0.0012$ in all six; holdout RMSE spans 3.6%. One pool and five pools are indistinguishable. ► [T1_model_summary](#) ► [F6b_skill_vs_complexity](#)

7.1.0.2 The ordering does not follow structure at all. Best and worst holdout RMSE belong to a five-pool and a three-pool model.

7.2 Out of sample, structure is nearly everything

7.2.0.1 The forecast diverges where the calibration agreed. ► [F4_initialization](#) (panel b)

7.2.0.2 Climate sensitivity differs $\sim 2\times$ across the family. Equilibrium SOC change at $+2^\circ\text{C}$: Yasso07 -21.7% , Yasso20 -13.7% , Yasso15 -9.6% . Between-model range is $3.1\times$ the mean within-model band. ► [climate_sensitivity_sweep](#)

Measured on an earlier run and as a *demonstration* (step change, single-exponential on bulk MRT). Must be recomputed on the current calibration before it goes in a figure.

7.3 The applied consequence

7.3.0.1 Model choice is a forecast decision, not a fit decision. Selecting an inventory model on calibration skill selects on a quantity that carries almost no information about what the model will project. This is the practical message of the paper.

7.3.0.2 So the deliverable is an ensemble, not a winner. An estimate plus its structural uncertainty.

Part VI

Act VI — Thread B, the new centre

8 The calibration wants more input *and* faster kinetics

This is the beat the old note did not have. Previously “inputs are the binding uncertainty”. Now something sharper and more falsifiable: the published parameterisations appear to *underestimate both* the litter reaching the soil and the rate at which it decomposes — and that is a statement about *where on the ridge* the truth sits, which has consequences.

8.1 Inputs

8.1.0.1 The multiplier has a physical referent. $\sigma_{\text{input}} \times \bar{J}$ is an effective litter flux: 3.19–4.34 tC ha $^{-1}$ yr $^{-1}$ across the six, i.e. 18–61% above the national total including understorey (~ 2.7). ► [SX_input_vs_literature](#)

8.1.0.2 Independent corroboration exists. Liski 2006 — same country, period, model family and inventory — implies a comparable correction. ► [F1_inputs](#)

8.2 Kinetics

8.2.0.1 Turnover is displaced from published values. Intrinsic MRT 24.6 / 27.1 / 22.6 yr (Yasso07/15/20) against published points 33.5 / 30.4 / 19.0. ► [F12_mrt_yasso](#)

8.2.0.2 Not one phenomenon. Yasso07’s whole gap is its *single* climate modifier (MRT = MRT_{ref}/ξ exactly); Yasso15/20 carry three pool-specific modifiers and cannot use that lever. Do not discuss the family’s gaps as one.

8.3 Why these are ONE claim, not two

8.3.0.1 The level pins only the product. $\text{level} \sim \text{MRT} \times \sigma_{\text{input}} \times \bar{J}$. Faster-with-more and slower-with-less fit identically; the posterior rides that ridge (Yasso07/15/20 corr $-0.73/-0.68/-0.64$; the simple models stronger still). ▶ [F14_mrt_ridge](#) ▶ [S13_mrt_ridge_benchmark](#)

The discipline this imposes. “Inputs too low” and “turnover too slow” are two coordinates of *one* displacement. Report it once. Claiming both as independent findings would double-count, and CLAUDE.md flags this in three places.

8.3.0.2 So what *is* the claim? That the ensemble’s best-fitting position on the ridge lies at higher input and faster turnover than the published parameterisations sit. That is a single, well-posed statement about location — and it is testable, because the two ends of the ridge are *not* equivalent in everything.

8.4 The independent witness (not on the ridge)

8.4.0.1 The models leave an input-shaped signal in their residuals. They explain $\sim 2\%$ of SOC variance; a random forest explains 44–65% of what they leave. ▶ [F11_rf_heatmap](#)

8.4.0.2 And the covariate is the same in all six. Stand basal area — a productivity proxy, not a kinetic one. This does not ride the ridge, so it is genuinely separate evidence that the limitation is on the input side.

Part VII

Act VII — Consequences, and how to test them

9 If the ensemble sits elsewhere on the ridge, what changes?

The scientific core of the next phase. A displacement that changed nothing observable would be bookkeeping. The bet is that it changes the *climate response* — which would make the position on the ridge an empirically decidable question rather than a prior-driven one.

9.1 The hypothesis

9.1.0.1 Equal fit, unequal reactivity. Two calibrations indistinguishable in fit, sitting at different points on the $\text{MRT} \times \sigma_{\text{input}}$ ridge, project *different* sensitivity to warming.

9.1.0.2 Why it should be so. Temperature acts on the *rate constants*. A faster-turnover solution has more of its carbon in pools that feel ξ on a decadal timescale, so the same warming moves more carbon. The input side responds through productivity instead — a different pathway, and one held fixed in these projections.

9.1.0.3 Already partly visible. Fit explains only 10–18% of forecast spread: 82–90% of the projected divergence is among draws the data cannot rank.

9.2 The test can be run now

Arm A vs arm B is the two-calibration experiment. The σ_{input} anchor sensitivity produced exactly the required design — the same six models calibrated under two input assumptions, both on disk with chains:

	arm A (Liski, centre 1.27)	arm B (L&H, centre 1.08)	shift
σ_{input} (Yasso15)	1.619	1.538	−5%
MRT Yasso07	22.13	24.57	+11%
MRT Yasso15	25.60	27.11	+6%
MRT Yasso20	20.82	22.56	+8%
effective flux	3.39–4.73	3.19–4.34	lower
fit	—	<i>marginally better</i>	—

Two positions on the ridge, comparable fit, no new runs required. Apply the climate-sensitivity machinery to both and read off the difference.

9.2.0.1 Pre-registration, before looking. Structure predicts *where* the effect should appear: Yasso07 large (single ξ ties MRT and climate sensitivity together; corr −0.58), Yasso15/20 small (three pool-specific ξ decouple them; corr −0.10/−0.16).

9.2.0.2 What would falsify the bet. A uniform, negligible difference across all three Yassos. That would mean the ridge is degenerate in climate response too, and the position on it is *not* empirically decidable from projection — a real result, and one worth reporting.

★ TO BUILD: F15 — climate reactivity, arm A vs arm B, per model, posterior bands

9.3 Stronger versions, if the cheap test pays off

9.3.0.1 Forward runs rather than a step response. Replace the single-exponential demonstration with actual forward integration under a warming scenario.

9.3.0.2 Deliberate ridge-walking. Calibrate under a restrictive and a permissive σ_{input} prior — wider separation than the anchor arms give.

9.3.0.3 Break the degeneracy from outside. Radiocarbon constrains turnover independently of input magnitude. The degeneracy is a property of the observation design, not of nature. Worth stating even if we cannot do it here.

The projections currently contain no climate change — the last twenty observed years are recycled. Everything above is therefore about *sensitivity*, computed separately, and the forecast figures are disequilibrium relaxation under stationary climate. The paper must say so.

Part VIII

Act VIII — Thread C

10 What the data can and cannot constrain

10.0.0.1 Convergence is not identifiability. $R\text{-hat} \leq 1.008$ everywhere, yet the transfer fractions are prior-pinned, ridge-correlated and carry < 1 nat. A clean sampler on a flat ridge. ► [F8b_fraction_ridges](#) ► [T2_fraction_ridges](#)

10.0.0.2 The data pin a level, and little else. Two observations per plot; no cross-plot skill. ► [F6_obs_vs_pred_holdout](#)

10.0.0.3 Report it, do not fix it. Non-identifiability here is a property of the observation design and belongs in the results.

Part IX

Act IX — Landing

11 The policy deliverable

11.0.0.1 An estimate plus its irreducible structural uncertainty. Not a single number, and not a selected winner.

11.0.0.2 Aggregate, not plot-level. These models capture the national trajectory; they do not rank plots. Say it early — otherwise “captures the dynamic” and “ $R^2 \approx 0$ ” read as a contradiction.

11.0.0.3 Maps as the landing figure. ► [NextGenC ensemble maps](#)

Part X

Act X — Outlook

12 Where the recoverable signal hides

12.0.0.1 Historical inputs. The deepest uncertainty; σ_{init} stands in for a century.

12.0.0.2 Understorey and belowground. The named gap in the litter product.

12.0.0.3 Stratified inputs. The fraction rises strongly south to north; one national multiplier cannot carry it.

12.0.0.4 Radiocarbon. The clean external constraint on turnover.

Part XI

Appendix

13 Figure inventory

13.0.0.1 Exist, need rebuilding on arm B. F1, F3, F4, F6, F6b, F11, F12, F14, S13, T1, T2 — all PNGs currently predate the current calibration.

13.0.0.2 Exist and are current. F5 (rebuilt 2026-09-02).

13.0.0.3 To build. F3b (shape groups), F15 (climate reactivity, arm A vs arm B), the variance partition schematic (optional).